Top 100
Global
Innovator
for 10 years

# DEXA

## DATA EXPERT AGENT

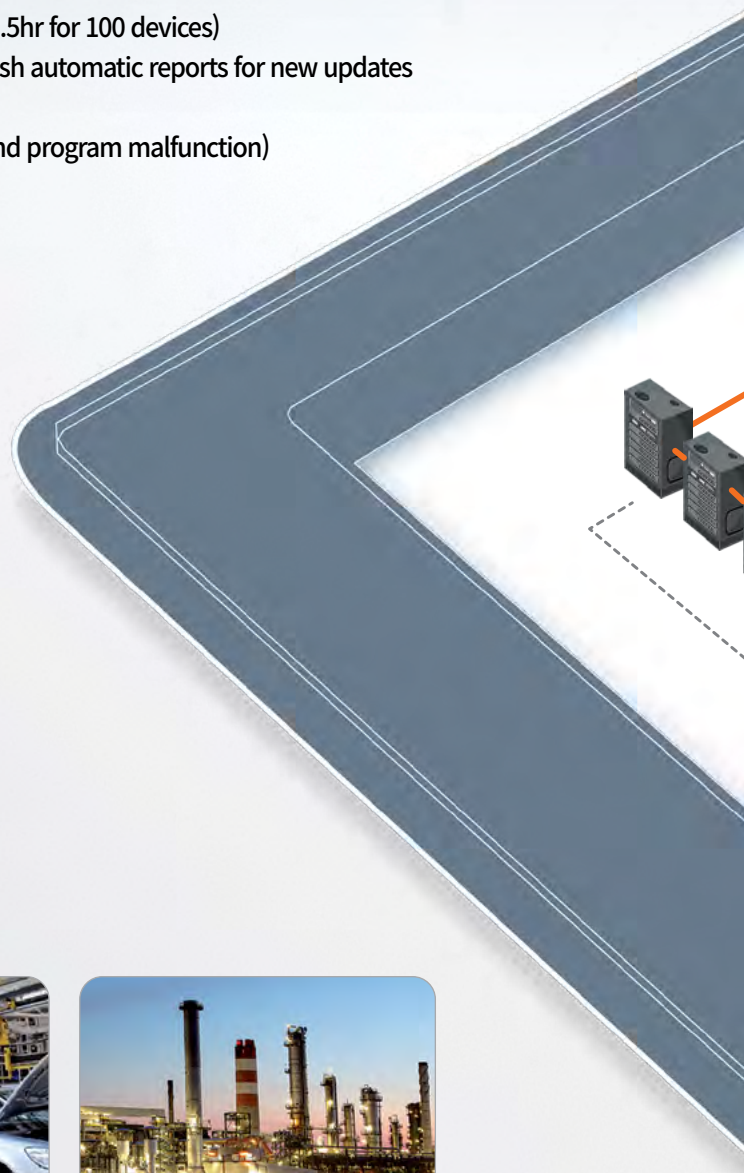
LS ELECTRIC

# DEXA
DATA EXPERT AGENT

## Essential for all smart factory process to store and backup engineering data

**DEXA provides outstanding performance in various network diagnosis, monitoring and backup functions with user-friendly interface and convenient operations.**

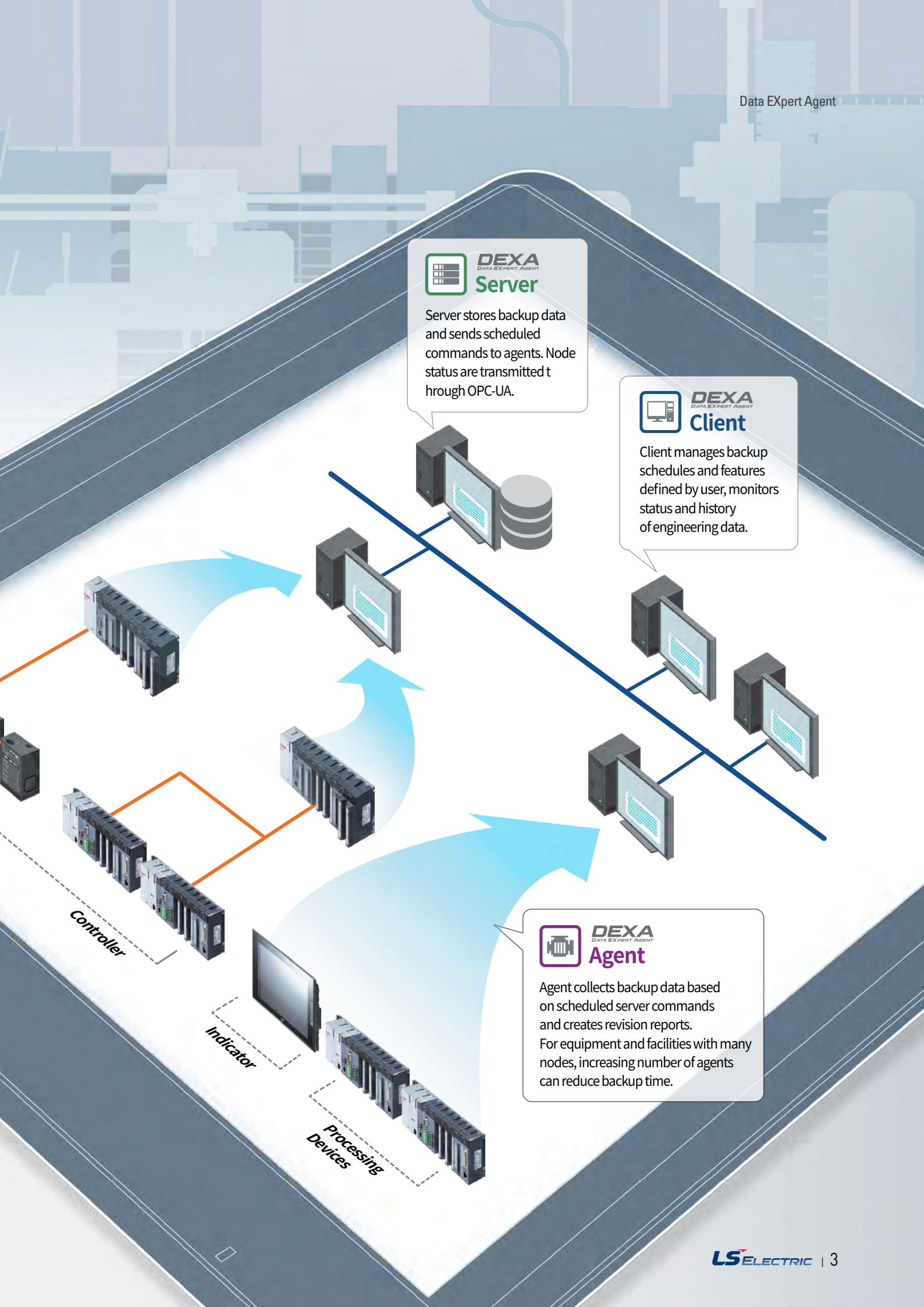
- Reduced backup time (less than 1hr for 60 devices, less than 1.5hr for 100 devices)
- Cross-check engineering data (program, parameter) and publish automatic reports for new updates
- Asset health status and real-time monitoring dashboard
- Log based detection for backup errors (network connection and program malfunction)
- Integrated management of customer engineering data



DEXA DATA EXPERT AGENT Server

DEXA DATA EXPERT AGENT Client

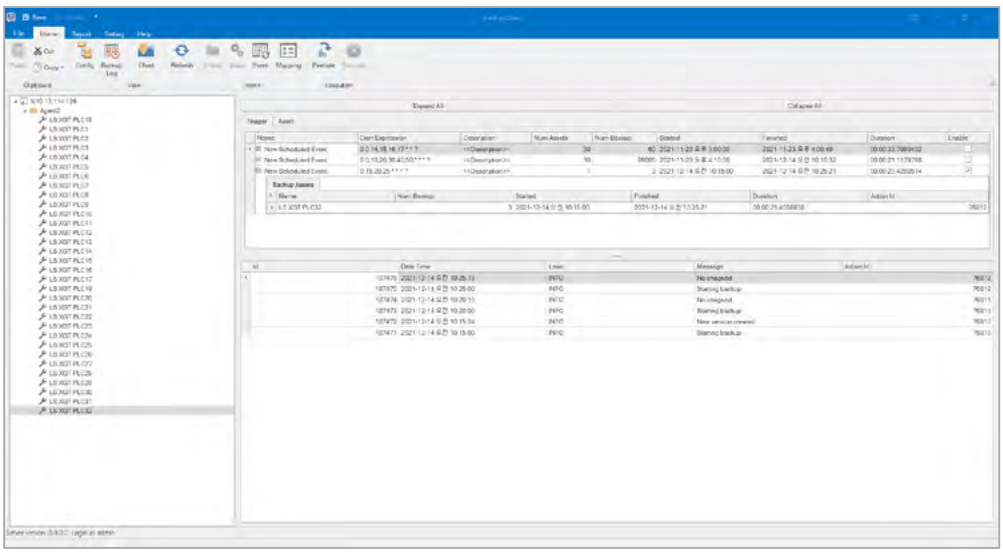
DEXA DATA EXPERT AGENT Agent


DEXA
DATA EXPERT AGENT
Server
Server stores backup data and sends scheduled commands to agents. Node status are transmitted t hrough OPC-UA.
DEXA
DATA EXPERT AGENT
Client
Client manages backup schedules and features defined by user, monitors status and history of engineering data.
DEXA
DATA EXPERT AGENT
Agent
Agent collects backup data based on scheduled server commands and creates revision reports.
For equipment and facilities with many nodes, increasing number of agents can reduce backup time.
Controller
Indicator
Processing Devices

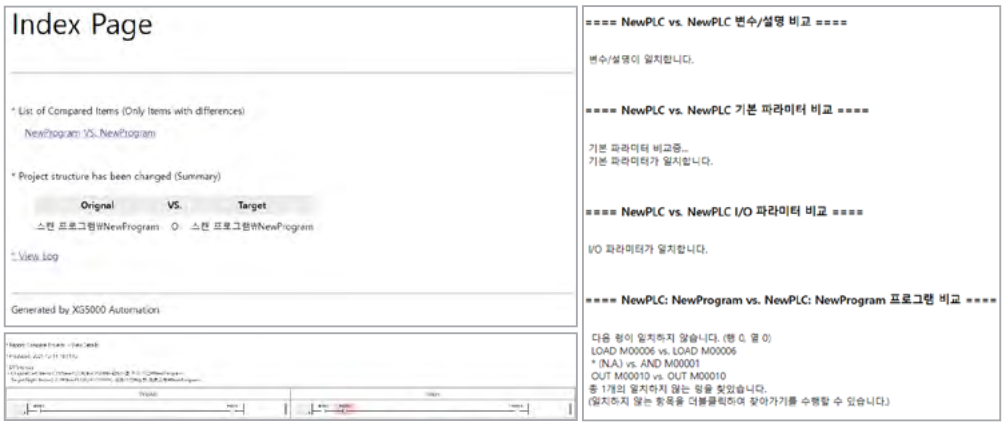
# Advantages

## Advantages

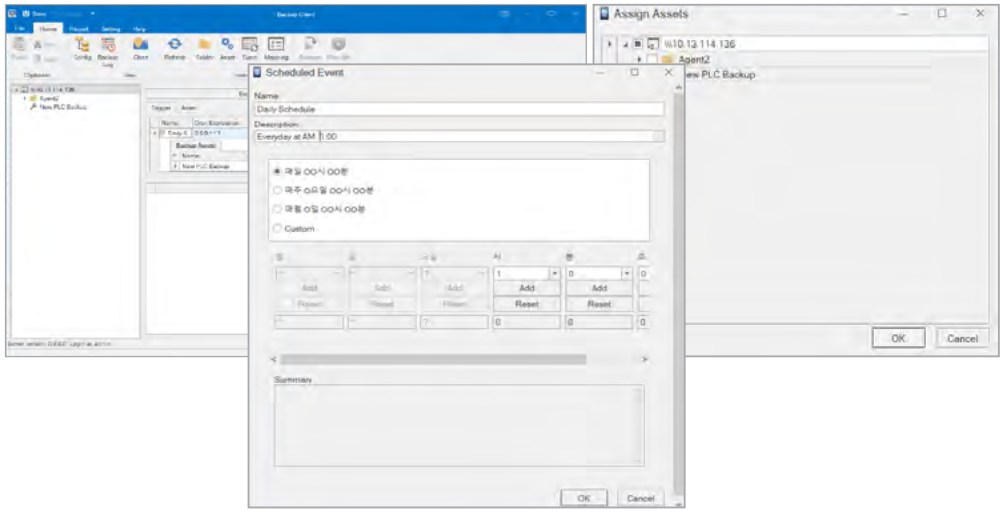
User-friendly interface with ribbon UI and directory structure



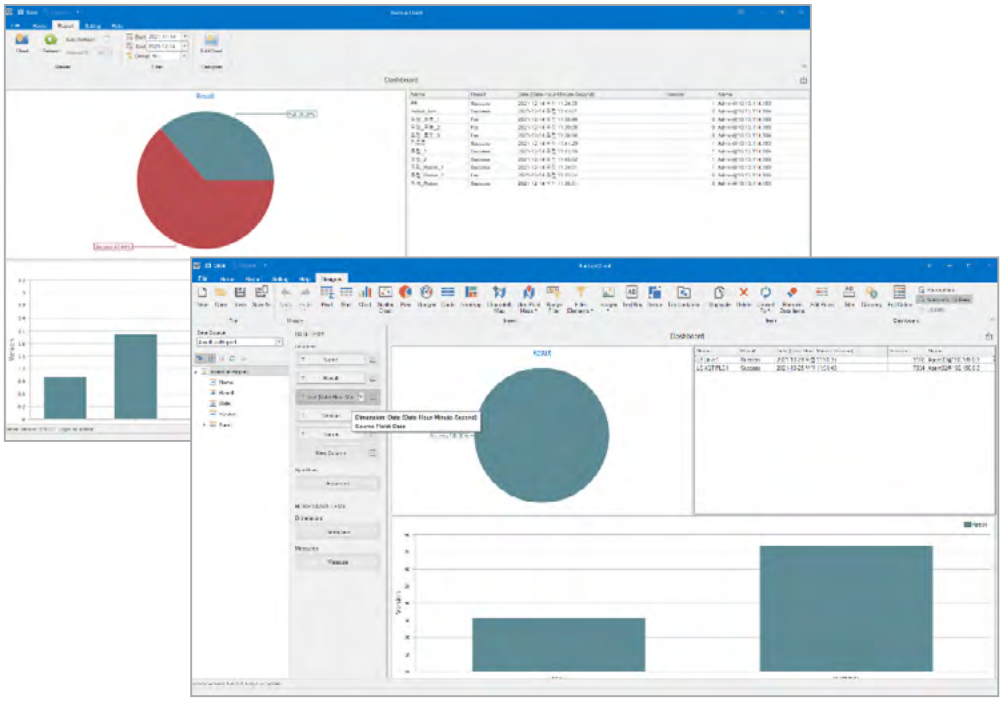
Cross-check engineering data (program, parameter, drawing)




Easy to schedule events for multiple devices

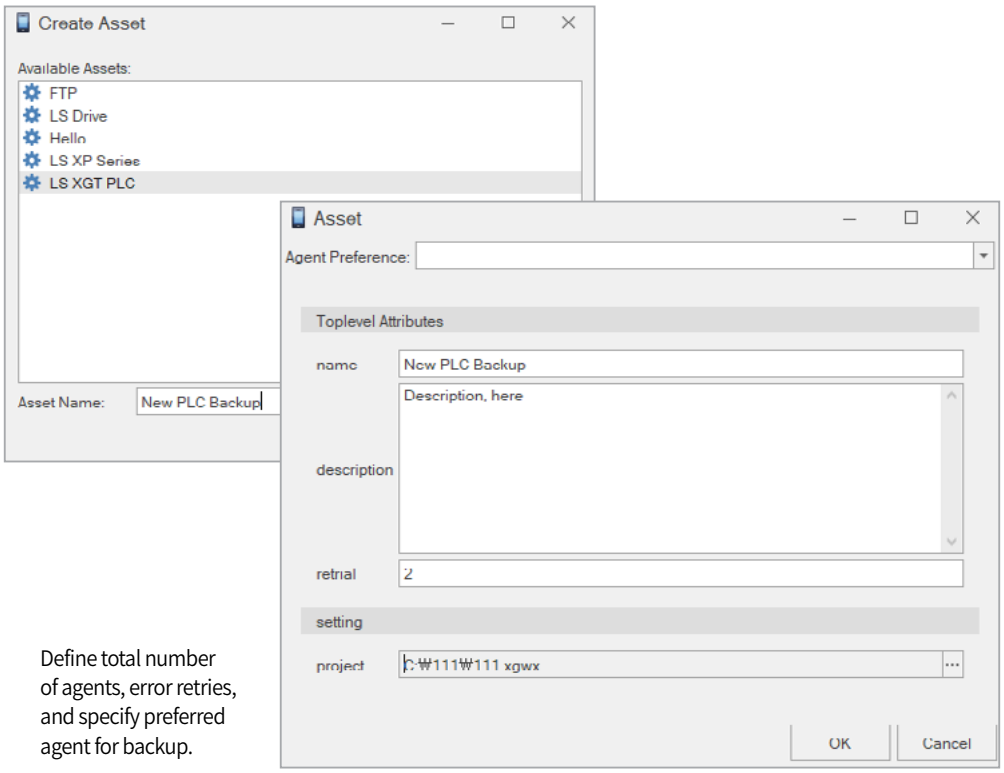


### Customize dashboard contents



## Settings

### Identical UX/UI for PLC, Drive, HMI and FTP

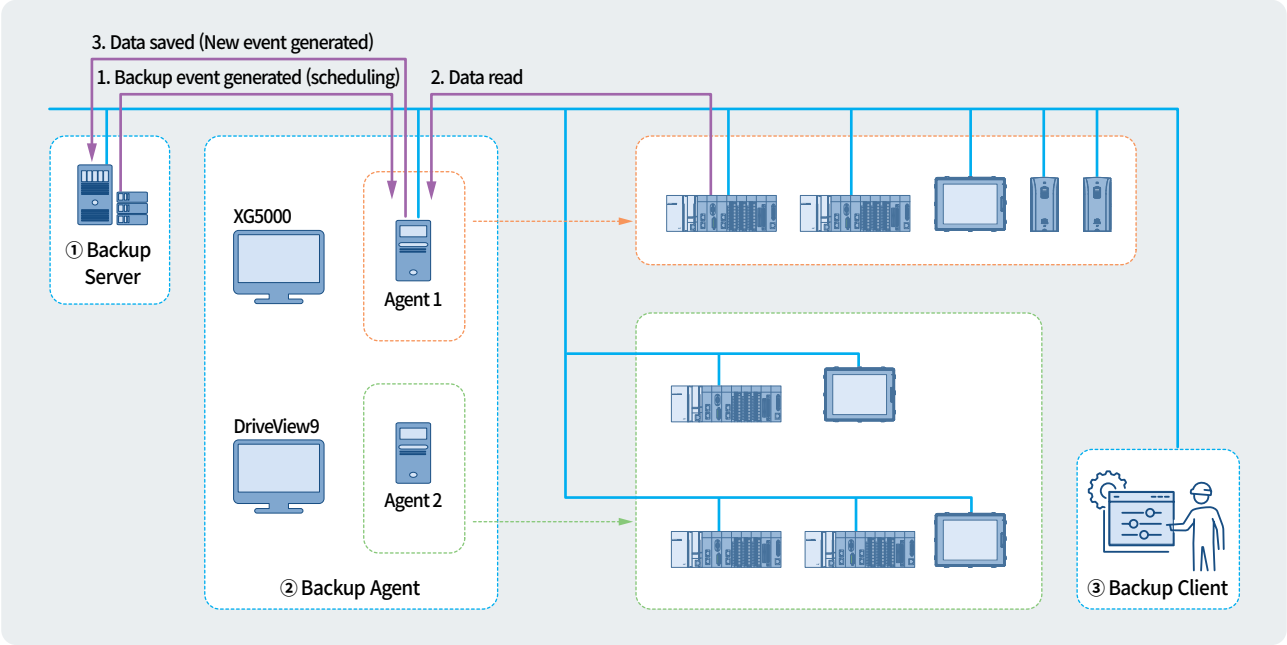



Define total number of agents, error retries, and specify preferred agent for backup.

## System Layout

**DEXA system is designed to store and retrieve all engineering data from the central server.**

- Simple Layout: Server PC (①Backup Server + ②Backup Agent) and PC(③Backup Client)
- Extended Layout: Server PC (①Backup Server), agent PC (②Backup Agent 1…N) and client PC(③Backup Client)




| DEXA | Function |
|---|---|
| ① Backup Server | Data service, storage agent management and scheduling |
| ② Backup Agent | Backup execution and report creation |
| ③ Backup Client | Backup setup and dashboard inspection |

| Devices | Models | Connection |
|---|---|---|
| PLC | XGT Series | CPU-embedded Ethernet or Ethernet communication cards (XGL-EFMTB, XBL-EFMT) |
| Drive | iS7, S100, G100, iP5A | Ethernet option card access |
| HMI | iXP, iXP2 | Web servers |

## Hardware Specifications

| System Layout | Function | Specifications | Remark |
|---|---|---|---|
| ① Backup Server | Data service, storage agent management and scheduling | Windows 10, Windows Server R2<br>.NET Framework 4.8<br>MS SQL Sever<br>CPU 2GHz or higher, RAM 8GB or more, SSD 512GB or more | Can be used with SQLite if not equipped with MS SQL Server |
| ② Backup Agent | Backup execution and report creation | Windows 10<br>.NET Framework 4.8<br>XG5000 V4.60 or more<br>DriveView 9 1.2.0 or more<br>CPU 1GHz or higher, RAM 4GB or more, SSD 256GB or more | Engineering S/W installed (Software version managed automatically) |
| ③ Backup Client | Backup setup and dashboard inspection | Windows 10<br>.NET Framework 4.0<br>CPU 1GHz or higher, RAM 4GB or more, SSD 256GB or more | Multiple clients provided. Exclusive access achieved with edit lock feature. |

## Functions

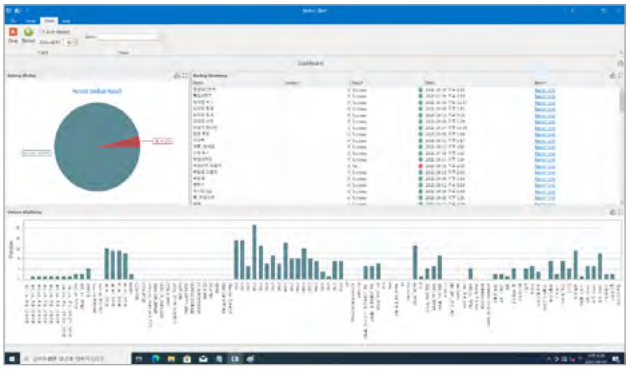
| Function | Devices | | | | Note |
|---|---|---|---|---|---|
| | PLC | Drive | HMI | FTP | |
| Schedule Backup | O | O | O | O | Sets backups on yearly/monthly/weekly/daily bases |
| Comparison Report Output | O | O | X | X | Prints detailed report comparing revisions |
| Backup Log Display | O | O | O | O | Prints backup log and error reports |
| Version Info Display | O | O | O | O | Version identification for revised items |
| Version Download | O | O | O | O | Downloads a specific version of engineering data and submits comparison reports |
| Configuration Management | O | O | O | O | Folder management for individual devices |
| Change Notifications | O | O | O | O | Reports changes in each folder (submitted via e-mail) |
| User Access Control | O | O | O | O | Manages user access to each folder |
| Storage Control | O | O | O | O | Supports saving on local disks, FTP and cloud storage |
| Keeping Up to Date | O | O | O | O | Manages a set number of latest versions |
| Agent S/W Synchronization | O | O | - | - | Autoupdates Agent softwares (XG5000, DriveView9) |
| Dashboard Display | O | O | O | O | Displays device backup status and summary |
| OPC-UA Node Display | O | O | O | O | Devices through OPC-UA |

## References

### LS ELECTRIC Cheonan Plant

- All backed up PLC projects are assigned with version numbers
- Latch area data can also be managed by backing up data memory
- Changes and revisions can be traced with comparison reports, which can be submiited via e-mail
- Dashboard can be configured by making conection to the EdgeHub using OPC UA

### Automotive Product Line



- All backed up PLC projects are assigned with version numbers
- Changes and revisions can be traced with comparison reports
- Personalized dashboard for each production line
- Supports multiple connections using built-in Ethernet port on the CPU (XGI-CPUUN)

## PC Line-ups

| Item | DEXA Server | | DEXA Agent |
|---|---|---|---|
| | Box type | Rackmount type | Mini |
| Appearance | | | |
| CPU | Intel Core i7-6700U | | Intel Celeron J1900 |
| OS | Window 10 IoT <Highend> / Window Server 2022 | | Window 10 IoT <Entry> |
| Memory | 16GB | | 8G |
| Storage | 512GB x 2EA | | 512GB |
| Communication Interface | 2×Gbe<br>5×USB<br>5×RS232/422/485 | | 2×Gbe<br>5×USB<br>5×RS232/422/485 |

## DEXA Line-ups

### SW Type

DEXA Server DEXA Client DEXA Agent

| Nodes | Basic type | Add-on |
|---|---|---|
| 25 | DEXA-A002-NP | DEXA-B002-NP |
| 50 | DEXA-A005-NP | DEXA-B005-NP |
| 100 | DEXA-A010-NP | DEXA-B010-NP |
| 200 | DEXA-A020-NP | DEXA-B020-NP |
| 500 | DEXA-A050-NP | DEXA-B050-NP |
| 1000 | DEXA-A100-NP | DEXA-B100-NP |

### Box Type / IPC Integrated

DEXA Server DEXA Client

| Nodes | Window 10 IoT | Window Server 2022 |
|---|---|---|
| 25 | DEXA-P002-NP-BI | DEXA-P002-NP-BS |
| 50 | DEXA-P005-NP-BI | DEXA-P005-NP-BS |
| 100 | DEXA-P010-NP-BI | DEXA-P010-NP-BS |
| 200 | DEXA-P020-NP-BI | DEXA-P020-NP-BS |
| 500 | DEXA-P050-NP-BI | DEXA-P050-NP-BS |
| 1000 | DEXA-P100-NP-BI | DEXA-P100-NP-BS |

### Rackmount Type / IPC Integrated

DEXA Server DEXA Client

| Nodes | Window 10 IoT | Window Server 2022 |
|---|---|---|
| 25 | DEXA-P002-NP-LI | DEXA-P002-NP-LS |
| 50 | DEXA-P005-NP-LI | DEXA-P005-NP-LS |
| 100 | DEXA-P010-NP-LI | DEXA-P010-NP-LS |
| 200 | DEXA-P020-NP-LI | DEXA-P020-NP-LS |
| 500 | DEXA-P050-NP-LI | DEXA-P050-NP-LS |
| 1000 | DEXA-P100-NP-LI | DEXA-P100-NP-LS |

### DEXA Agent PC

DEXA Agent

| Nodes | Window 10 IoT |
|---|---|
| - | DEXA-PAGT |

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